

ROUTING & SWITCHING ESSENTIALS I





COURSE STRUCTURE

Day 1: Ethernet Switching, VLANs, Routing, OSPF & EIGRP

Day 2: STP, EtherChannel, HSRP, DHCP, NAT & Data Center Architecture

Day 3: Networking in the Cloud Era — AWS VPC, EC2 & Security

Day 4: Cloud Network Services & Hybrid Connectivity

Day 5: High Availability, Troubleshooting & Network Management Platforms

AGENDA FOR DAY 1

1. CAM Table, MAC Learning & Port Security
2. VLANs: Configuration, Trunking & Design
3. Inter-VLAN Routing — ROAS & SVIs
4. Routing Fundamentals: RIB, FIB, AD & Longest Prefix Match
5. Static Routing
6. OSPF: Neighbors, DR/BDR, Cost & Troubleshooting
7. EIGRP Overview & OSPF vs EIGRP Comparison



ETHERNET SWITCHING FUNDAMENTALS

CAM TABLE & L2 FORWARDING

- Switches use a CAM (Content Addressable Memory) table to make every forwarding decision
- CAM Stores MAC addresses, port, VLAN
- Frame forwarding based on destination MAC
- Unknown destination → flooded out all ports except incoming
- Dynamic vs static MAC entries



Mac Adress	Port	VLAN
0000.0000.1111	1	10
0000.0000.2222	2	10
0000.0000.3333	7	20

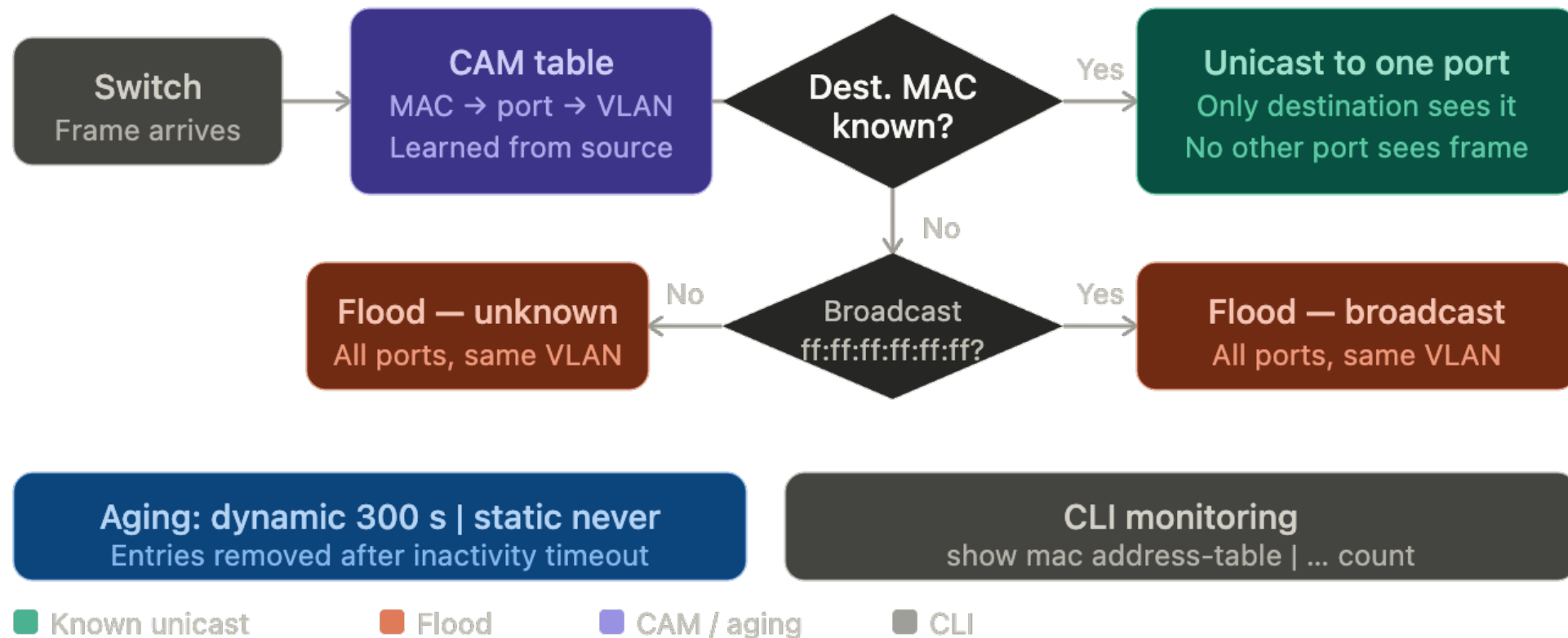
CAM TABLE & L2 FORWARDING

- Broadcast (ff:ff:ff:ff:ff:ff) → always flood to all ports in VLAN — never filtered
- Dynamic entries age out after 300 seconds; static entries never age out
- `show mac address-table` | `show mac address-table count` — view and monitor table



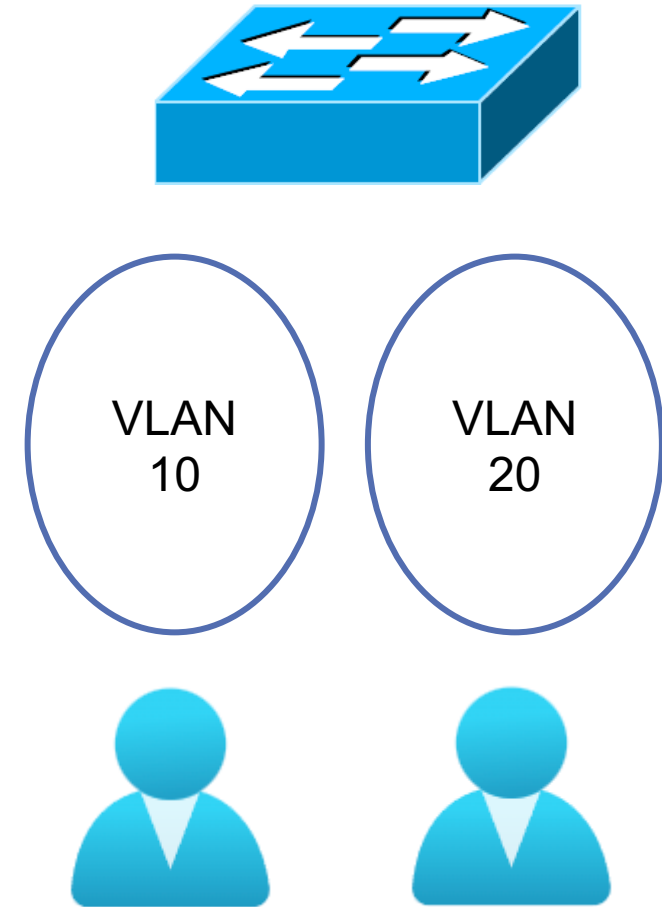
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CAM TABLE & L2 FORWARDING



VLAN CONCEPT & BENEFITS

- VLAN = Virtual LAN; logical segmentation of a switch
- Separates broadcast domains
- Increases security and efficiency
- Simplifies management and policy enforcement
- VLANs identified by 12-bit VLAN ID (0–4095, 1–4094 usable)



802.1Q TAGGING & NATIVE VLAN

- IEEE 802.1Q = VLAN tagging standard
- Adds 4-byte tag to Ethernet frame
- Contains VLAN ID and priority (CoS) bits
- Native VLAN = untagged traffic on trunk links
- Security best practice: avoid using VLAN 1 as native



802.1Q TAGGING & NATIVE VLAN

Before tagging

Plain Ethernet — access port or native VLAN on trunk



802.1Q inserts 4-byte tag after Src MAC



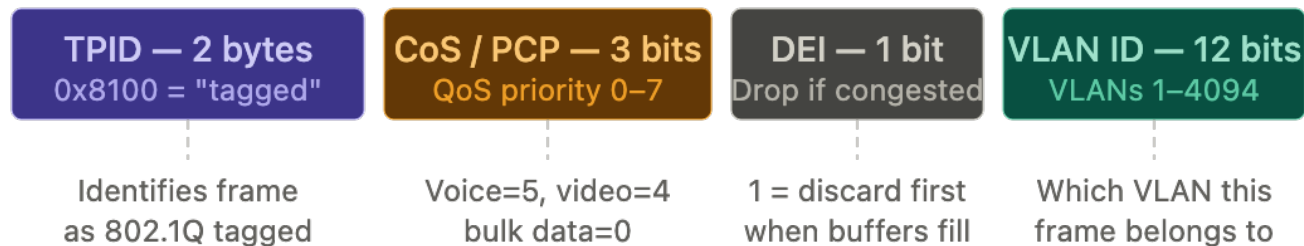
After tagging

Tagged Ethernet frame — carried on trunk links between switches



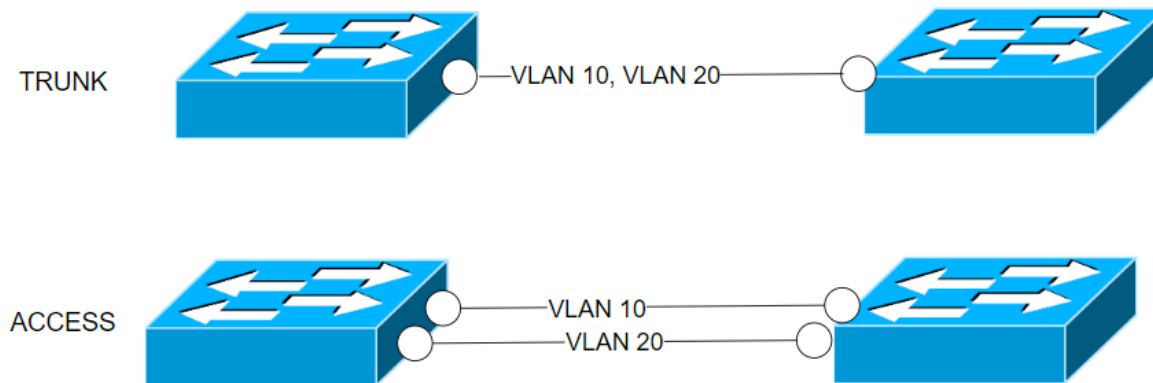
Tag breakdown

The 4 bytes of the 802.1Q tag contain four fields



Native VLAN exception: frames in the native VLAN travel untagged — no 802.1Q tag inserted

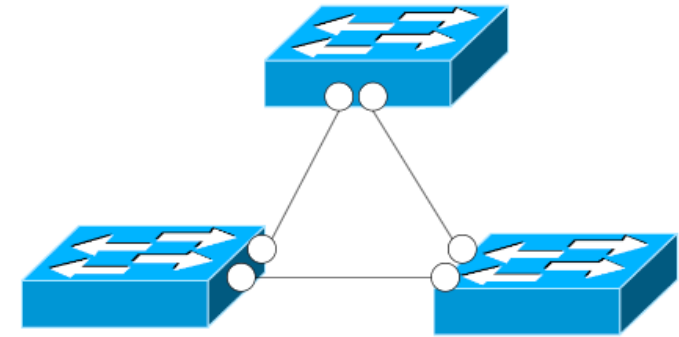
ACCESS VS TRUNK PORTS



- Access port: carries traffic for one VLAN
- Trunk port: carries multiple VLANs (tagged traffic)
- Dynamic Trunking Protocol (DTP) negotiates trunks (Cisco)
- Common commands:
switchport mode
access/trunk

MULTI-SWITCH VLAN PROPAGATION

- VLANs can span multiple switches via trunks
- VLAN database synchronized using VTP (Cisco)
- Transparent vs Server/Client modes
- Always verify VLAN consistency across switches



POP QUIZ:

What is the purpose of the CAM table in a network switch?

- A. To store routing information
- B. To store MAC addresses and associated ports
- C. To store IP-to-MAC mappings
- D. To filter broadcast traffic

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POP QUIZ:

What happens when a switch receives a frame with an unknown destination MAC address?

- A. It drops the frame
- B. It sends it to the default gateway
- C. It floods it out all ports except the incoming one
- D. It adds it to the CAM table

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Which type of MAC entry in a switch is manually configured by an administrator?

- A. Dynamic
- B. Static
- C. Temporary
- D. Broadcast



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- A. To encrypt the frame payload for security
- B. To identify the VLAN the frame belongs to
- C. To increase the MTU beyond 1500 bytes
- D. To specify the destination IP address



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POP QUIZ:

Which VLAN's traffic is sent UNTAGGED on an 802.1Q trunk?

- A. VLAN 1 always
- B. The highest VLAN ID configured
- C. The native VLAN
- D. The management VLAN



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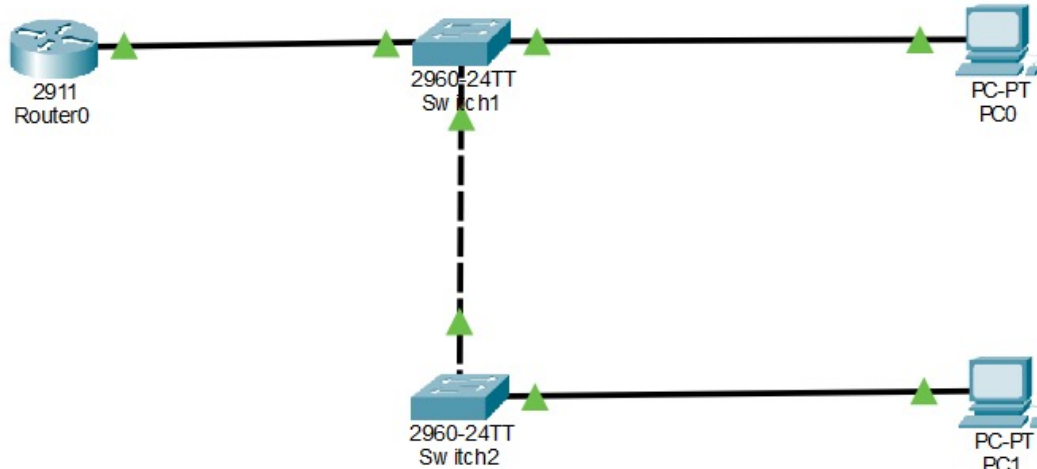
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INTER-VLAN ROUTING & DESIGN

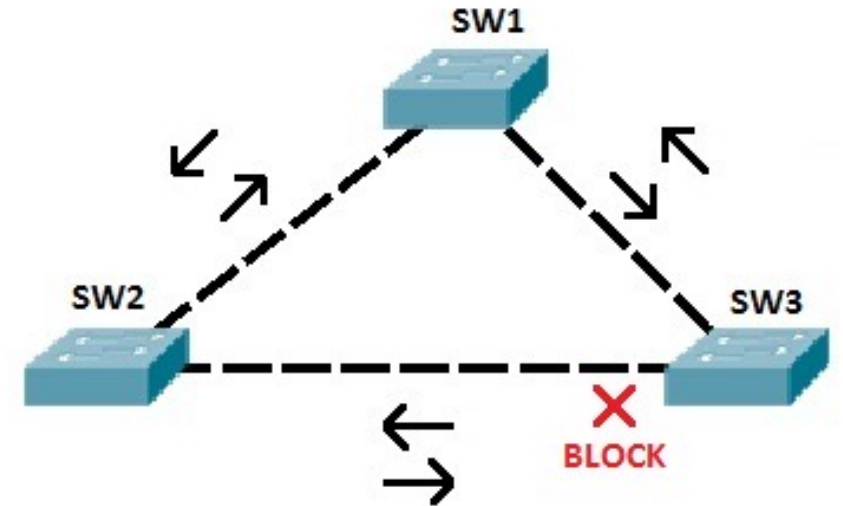
INTER-VLAN ROUTING MODELS



- Router-on-a-Stick (ROAS)
- One physical link, subinterfaces per VLAN
- Layer 3 Switch (SVI-based)
(Switched Virtual Interface) virtual Layer 3 interface
- Inter-VLAN routing in hardware
- L3 switches = faster, scalable solution

STP BASICS: ROOT BRIDGE, PORT ROLES & STATES

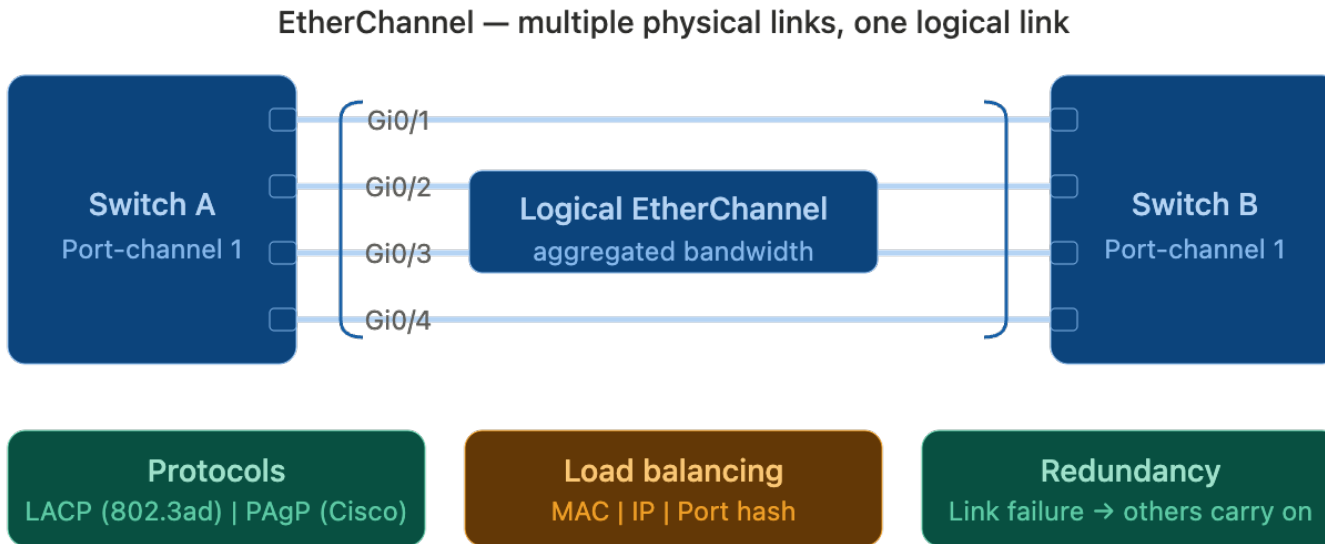
- Spanning Tree Protocol prevents loops
- Elects Root Bridge (lowest Bridge ID)
- Port roles: Root, Designated, Alternate
- Port states: Blocking, Listening, Learning, Forwarding



STP ENHANCEMENTS: PORTFAST & BPDU GUARD

- PortFast: immediate forwarding on access ports
- BPDU (Bridge Protocol Data Unit) Guard: disables port if BPDU received
- Safety mechanism for access ports where you do not expect to receive a BPDU
- Enhancements improve stability and security

ETHERCHANNEL / LACP OVERVIEW



- Combines multiple physical links into one logical link
- Increases bandwidth and redundancy
- Protocols: PAgP (Cisco), LACP (802.3ad standard)
- Load balancing based on MAC/IP/Port

VERIFYING VLANS & TRUNKS

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/2, Fa0/3, Fa0/4, Fa0/5 Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Gig0/1, Gig0/2
10	HR	active	Fa0/1
20	FINANCE	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

- show vlan brief – VLANs and ports
- show interfaces trunk – trunk status and allowed VLANs
- show interfaces switchport – mode and VLAN info

VERIFYING STP STATE & ROOT ELECTION

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address    0001.6387.A8D3
             Cost        19
             Port        23(FastEthernet0/23)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address    0009.7C84.A89C
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface                Role Sts Cost          Prio.Nbr Type
-----
Fa0/24                   Desg FWD 19           128.24  P2p
Fa0/23                   Root FWD 19           128.23  P2p

VLAN0010
  Spanning tree enabled protocol ieee
  Root ID    Priority    32778
             Address    0001.6387.A8D3
             Cost        19
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             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
```

- show spanning-tree – displays STP status
- Identify Root Bridge and port roles
- Check timers and priority values
- Common root election issue: unintentional root bridge

VERIFYING ETHERCHANNEL (PAGP/LACP)

- show etherchannel summary
- show interfaces port-channel
- Verify protocol, mode, and member links
- Check load balancing algorithm

TROUBLESHOOTING VLAN MISMATCHES

```
Switch#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme  Capability  Platform  Port ID
Router         Fas 0/24      159      R           C2900      Gig 0/0
Router         Fas 0/24      159      R           C2900      Gig 0/0.10
Router         Fas 0/24      159      R           C2900      Gig 0/0.20
Switch         Fas 0/23      159      S           2960       Fas 0/23
Switch#
```

- Check VLAN assignments on both sides
- Ensure trunk allowed VLANs match
- Review VTP mode/version consistency
- Use show cdp neighbor detail for mismatch hints

TROUBLESHOOTING STP LOOPS

- Identify duplicated frames and MAC flapping
 - Use show mac address-table for flapping ports
 - Check BPDU Guard logs
- Verify redundant connections properly blocked

```
Switch#show mac address-table  
Mac Address Table
```

Vlan	Mac Address	Type	Ports
1	0002.1701.3417	DYNAMIC	Fa0/23
1	0090.21bc.7a01	DYNAMIC	Fa0/24
10	0002.1701.3417	DYNAMIC	Fa0/23
10	0090.21bc.7a01	DYNAMIC	Fa0/24
20	0002.1701.3417	DYNAMIC	Fa0/23
20	0090.21bc.7a01	DYNAMIC	Fa0/24

DESIGN TIPS: VLAN NUMBERING & NAMING

- Use structured VLAN numbering (Dept/Function/Location)
- Example: VLAN 10–19 = HR, 20–29 = Finance
- Use descriptive names
- Reserve high VLAN IDs for future expansion



MGMT VLAN & OUT-OF-BAND IDEAS

- Dedicated management VLAN for switches
- Prevents control traffic mixing with user data
- Optional out-of-band (OOB) network via console or mgmt port
- Restrict access with ACLs

DOCUMENTATION MUST-HAVES FOR L2

- VLAN numbering and naming plan
- Trunk and EtherChannel diagrams
- STP root bridge roles
- Management VLAN design
- Verification and troubleshooting checklist



POP QUIZ:

What is the main benefit of implementing VLANs in a network?

- A. Increases broadcast traffic
- B. Combines multiple networks into one
- C. Segments broadcast domains for security and efficiency
- D. Simplifies routing configuration



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POP QUIZ:

Which type of port carries traffic for multiple VLANs?

- A. Access port
- B. Trunk port
- C. Hybrid port
- D. Core port



POP QUIZ:

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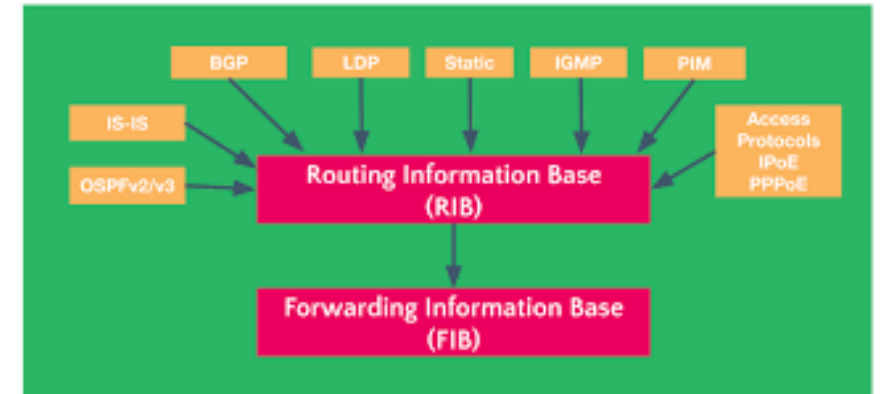
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ROUTING FUNDAMENTALS

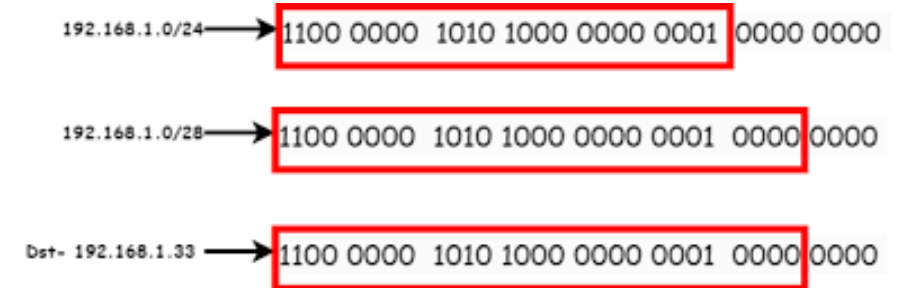
ROUTING FUNDAMENTALS

- RIB (Routing Information Base):
 - Stores all learned routes (static + dynamic)
- FIB (Forwarding Information Base):
 - Optimized routing table used for actual packet forwarding
- Relationship:
 - RIB → FIB (control plane → data plane)



LONGEST PREFIX MATCH

- Router selects the most specific route that matches the destination
- Example:
 - 10.0.0.0/8 → broad
 - 10.0.1.0/24 → more specific → preferred
- Ensures efficient routing decisions



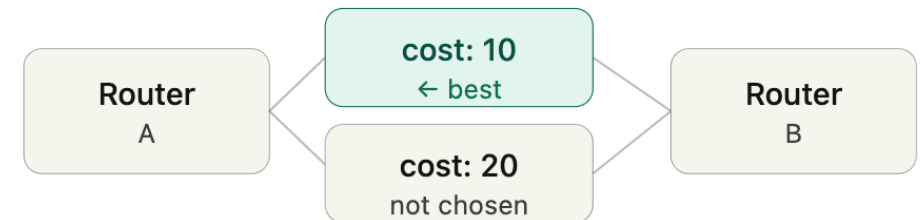
ADMINISTRATIVE DISTANCE & METRICS

- Administrative Distance (AD):
 - Trust level of a routing source
 - Direct = 0
 - Static = 1
 - EIGRP = 90
 - OSPF = 110
 - RIP = 120
- Metric:
 - Value used within a protocol to choose the best path

Protocol / source	AD
Direct connected	0
Static route	1
EIGRP	90
OSPF	110
RIP	120

lower AD = more trusted · overrides higher-AD sources to same destination

Within one protocol, metric decides the best path



STATIC ROUTING – PROS & CONS

-  Advantages:
 - Simple to configure
 - Predictable paths
 - Secure (no routing updates)
-  Disadvantages:
 - No automatic failover
 - Manual updates required
 - Poor scalability

OSPF AREA 0 (SINGLE-AREA OVERVIEW)

- OSPF: Open Shortest Path First = Link-State Protocol
- Organizes network into areas
- Area 0 = Backbone (mandatory core area)
- Single-area OSPF: All routers share one LSDB

OSPF NEIGHBORS, DR & BDR

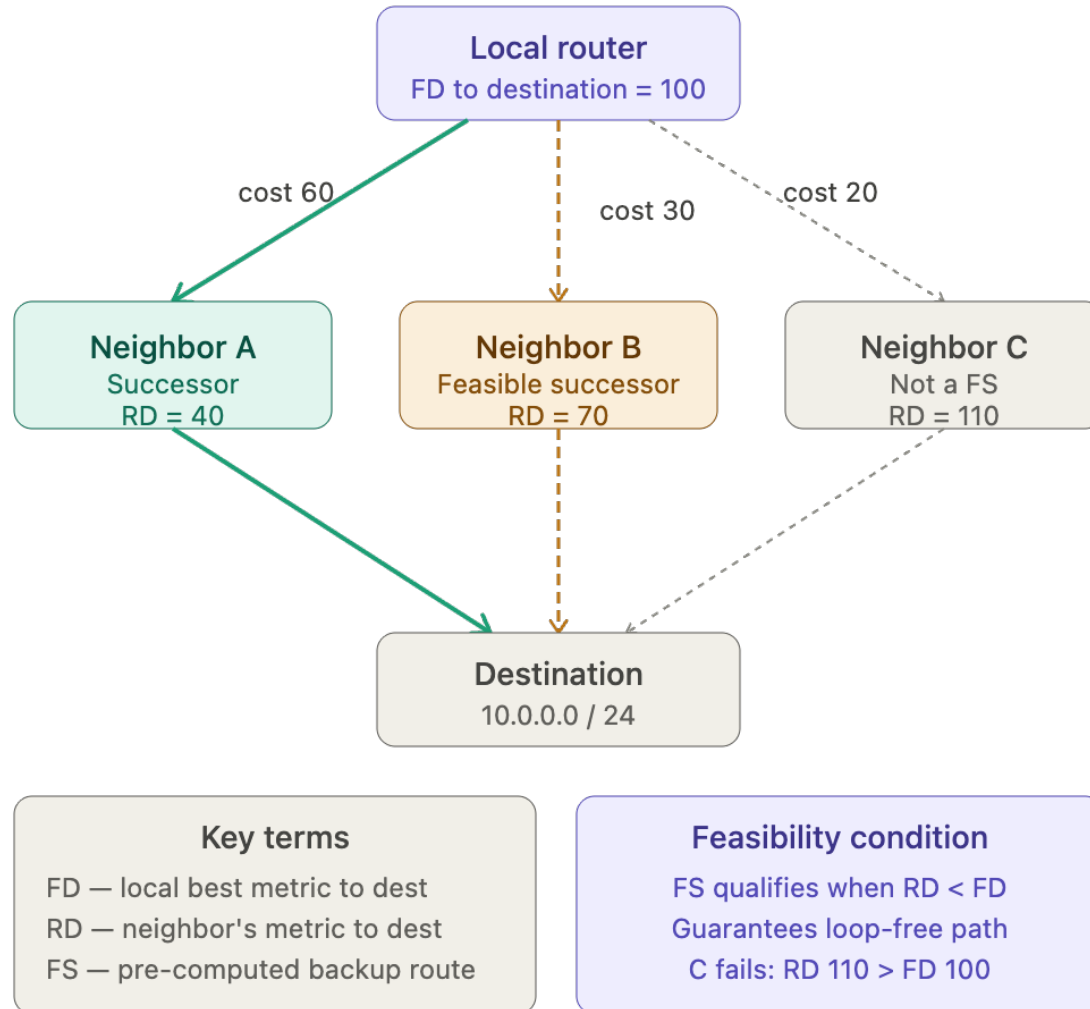
- OSPF routers form neighbor relationships via Hello packets
- On multi-access networks:
 - DR (Designated Router)
 - BDR (Backup DR)
 - Reduces LSA flooding on shared segments

OSPF COST & PATH CONTROL

- $\text{Cost} = \text{Reference Bandwidth} / \text{Interface Bandwidth}$
- Default reference = 100 Mbps
- Lower cost → preferred path
- Cost can be manually adjusted for path optimization

EIGRP: ENHANCED INTERIOR GATEWAY ROUTING PROTOCOL

DUAL algorithm — pre-computed failover
No waiting for reconvergence when a Feasible Successor exists

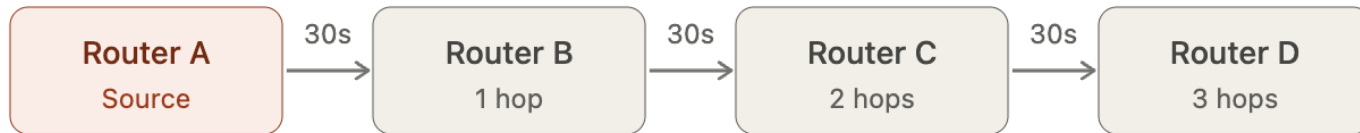


- Cisco-proprietary hybrid protocol (partially open standard RFC 7868)
- DUAL algorithm: pre-computes backup routes
- Near-instant failover
- AD: internal = 90 | external = 170 — EIGRP internal preferred over OSPF (110) for same prefix

RIP: ENHANCED INTERIOR GATEWAY ROUTING PROTOCOL

RIP — routing by rumour

Each router adds 1 hop and passes the full table to its neighbour



Routing table at B:

```
10.0.0.0 via A 1 hop
10.0.1.0 via A 2 hops
10.0.2.0 via A 3 hops
```

Max 15 hops — hop count 16 = unreachable

No awareness of link speed or bandwidth

- Distance-vector protocol: routers only know what neighbors report
- Updates broadcast every 30 seconds
- Metric is hop count only. No awareness of bandwidth or link speed
- Maximum 15 hops. Any destination beyond that is unreachable
- Legacy only, replaced by OSPF and EIGRP in modern networks

ROUTING PROTOCOL COMPARISON

Protocol	Type	AD	Metric	Scale	Convergence
Direct	Local interface	0	None	N/A	Instant
Static	Manual config	1	None	Small only	Manual update
EIGRP	Hybrid IGP	90 170 external	BW + Delay	Large	Very fast (DUAL)
OSPF v2/v3	Link-state IGP	110	Cost (ref BW)	Very large	Fast (SPF calc)
RIP v2	Distance-vector	120	Hop count (max 15)	Small only	Slow (30s timer)

Administrative distance — lower value wins



PASSIVE INTERFACE DESIGN

- Passive interface: Stops sending routing updates but still advertises its network
- Commonly used on LAN interfaces connected to end devices
- Increases security and reduces unnecessary routing traffic

POP QUIZ:

What is the relationship between RIB and FIB in routing?

- A. RIB is for control plane; FIB is for data plane forwarding
- B. RIB handles VLAN tagging
- C. FIB manages routing protocols
- D. They are identical tables



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POP QUIZ:

Which routing source has the lowest administrative distance?

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- B. Static route
- C. RIP
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POP QUIZ:

What rule determines which route is chosen when multiple matches exist?

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- C. Random selection
- D. Load balancing



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LAB 01: VLANS AND TRUNKING

- Goal: Segment network traffic using VLANs.
- Steps:
 - Create multiple VLANs on switches.
 - Configure access ports for end devices.
 - Set up trunk links between switches.
 - Verify VLAN isolation and trunk operation.

LAB 02: INTER-VLAN ROUTING

- Goal: Enable communication between VLANs.
- Steps:
 - Configure router-on-a-stick topology.
 - Create VLAN sub-interfaces on router.
 - Test inter-VLAN routing functionality.
 - Troubleshoot VLAN connectivity issues.

LAB 03: STP AND ETHERCHANNEL

- Goal: Implement redundancy and load balancing.
- Steps:
 - Configure Spanning Tree Protocol.
 - Create EtherChannel link aggregation.
 - Test failover scenarios.
 - Monitor STP topology changes

LAB 04: VLAN SECURITY AND BEST PRACTICES

- Goal: Harden VLAN and trunk configurations using essential Layer 2 security best practices.
- Steps:
 - Create secure VLANs including native VLAN (99) and parking lot VLAN (999).
 - Assign end-user VLANs (10 and 20) to PC ports.
 - Configure trunk Fa0/24 with static trunking and disabled DTP.
 - Move the native VLAN away from VLAN 1.
 - Apply parking lot VLAN + shutdown to unused ports.
 - Verify VLAN assignments and trunk security settings with show commands.
 - Test expected connectivity behavior (no inter-VLAN routing).